



GURU NANAK DEV ENGINEERING COLLEGE, LUDHIANA

(AN AUTONOMOUS COLLEGE U/S 2(F) & 12(B) OF UGC ACT - 1956)

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1

Dated _____ Class _____ Sec _____

University Roll. No. _____ Signature of Invigilator _____

Class Roll No. _____ Subject _____

Q. No.	1	2	3	4	5	6	7	Total Marks	Sig. of Examiner
Marks									

Ch 2 AC Circuits

2 Types of system

DC system

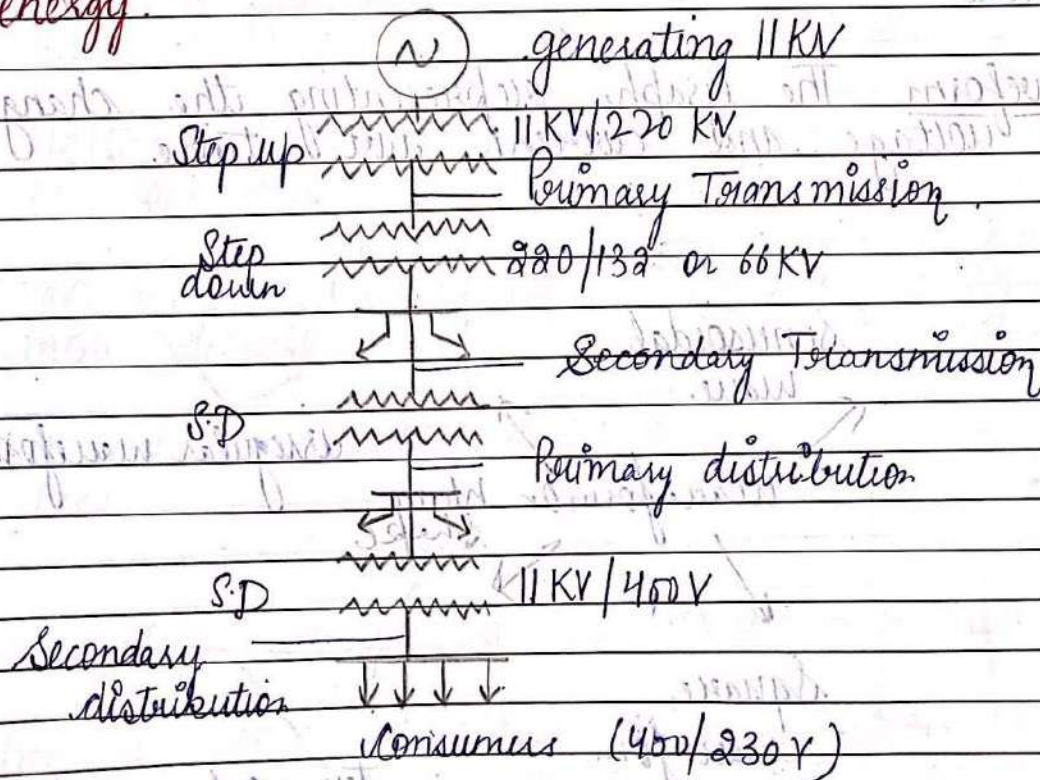
(Steady current flow & in one direction)

Application 1. electroplating
2. electronic circuit

AC system

(Current that changes the polarity and magnitude at regular intervals of time) app. use in generation, transmission & distribution of electrical energy.

Generation, Transmission & Distribution of Electrical Energy.



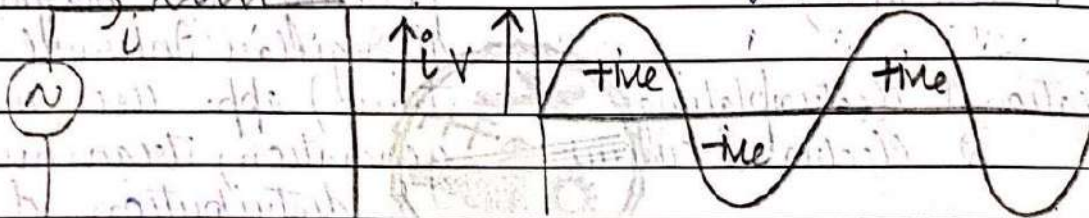
Advantages of AC system over DC system

1. The ac voltage can be stepped up & stepped down using transformers.
2. The ac motor are cheaper in cost, simple in construction and more efficient than dc motors.

Alternating Voltage & Current

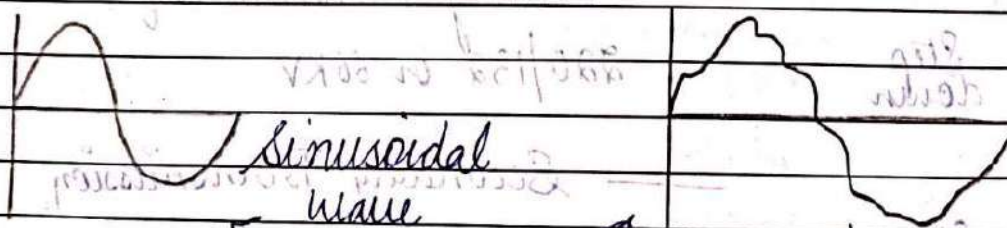
a. Alternating Voltage & Current

Alternating Voltage: A voltage that changes its polarity and magnitude at regular interval of time is called an alternating voltage.

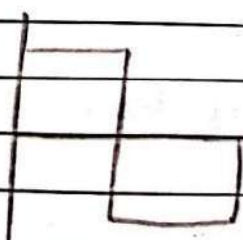


b) Alternating current: When an alternating voltage source is connected across load the current which flows through it is an alternating current.

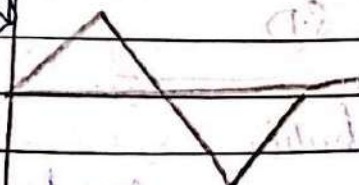
c) Waveform The graph representing the change in voltage and current with time.



Waveform or wave shape



Square waveform



Sinusoidal Alternating quantities

The voltage & current which varies according to sine of angle θ is known as sinusoidal alternating voltage or current.

The sinusoidal alternating quantities are used in all over the world.

1. As it produce less Iron & Cu losses in ac machines.
2. Offer less Interference to nearby communication system (telephone etc)

Difference between AC & DC

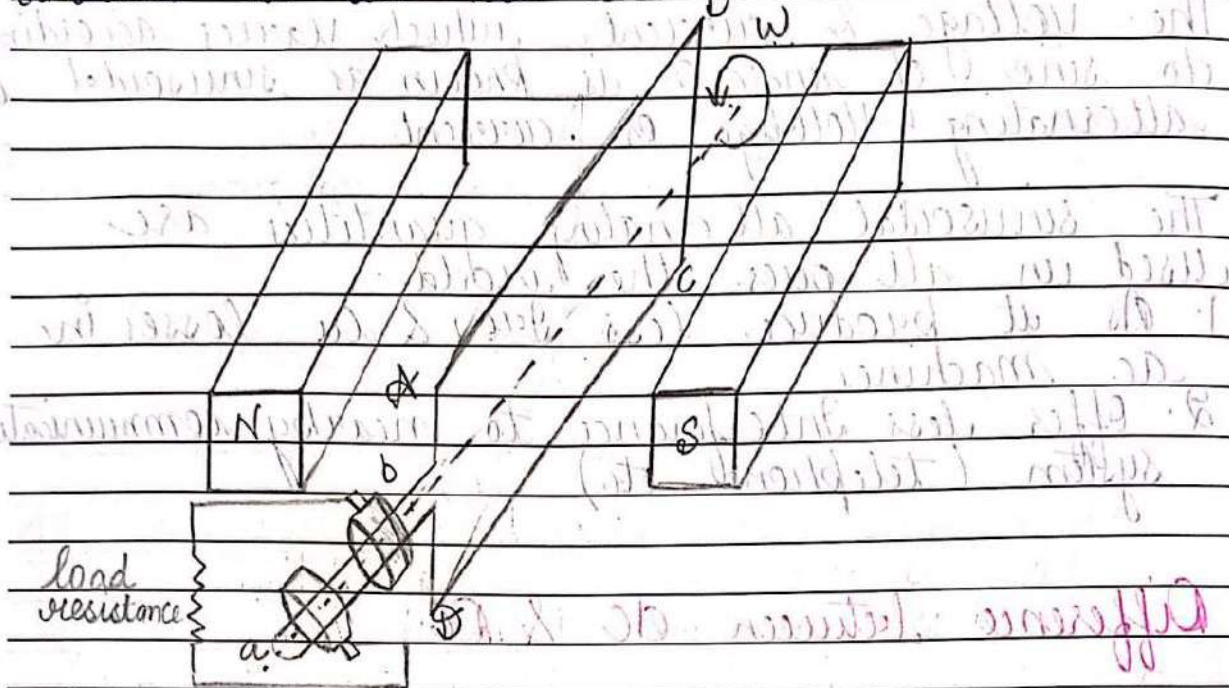
Alternating Current	Direct Current
---------------------	----------------

- | | |
|---|--|
| 1. Change polarity & magnitude | 1. flow only in one direction |
| 2. Can be generated at higher voltages | 2. DC can't be generated at high V because of commutation difficulties |
| 3. Cost of generation is less | 3. Cost of generation is more |
| 4. Voltage can be step up or step down easily | 4. Voltage can't be increased |
| 5. AC motor less costly & more durable | 5. more costly & less durable |
| 6. maintenance cost is less | 6. maintenance cost is high |

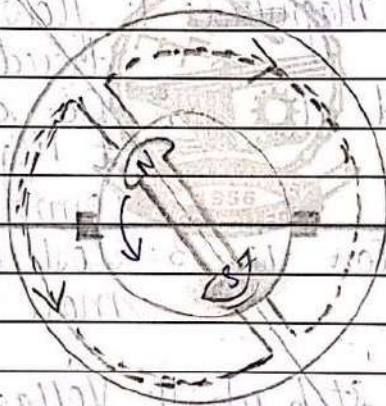
Generation of alternating Voltage & Current.

When a coil is placed in a uniform MF and start revolving in anticlock wise direction at

const. angular velocity (ω) per sec, an emf is induced in the coil side.

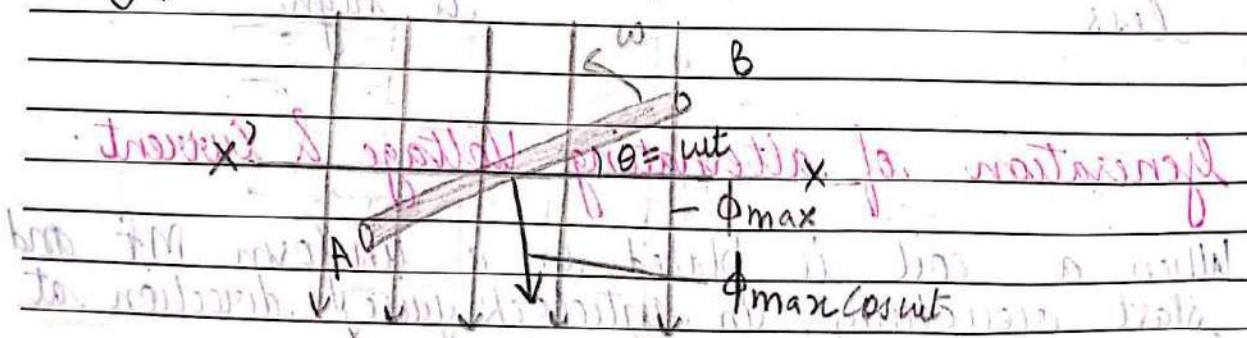


* Rotating coil in uniform M.F at constant speed used in small ac generator.



* Rotating a uniform M.F with a stationary coil at constant speed used in large AC generator.

Let the coil is in the position as shown in fig 3.



The angle through which the coil has rotated in t secs = ωt ($\theta = \omega t$)

The flux $\perp$ to the plane of coil = $\phi_{max} \cos \omega t$

instantaneous flux linkage = $N \phi_{max} \cos \omega t$
 (No. of turns on coil) (linking flux)

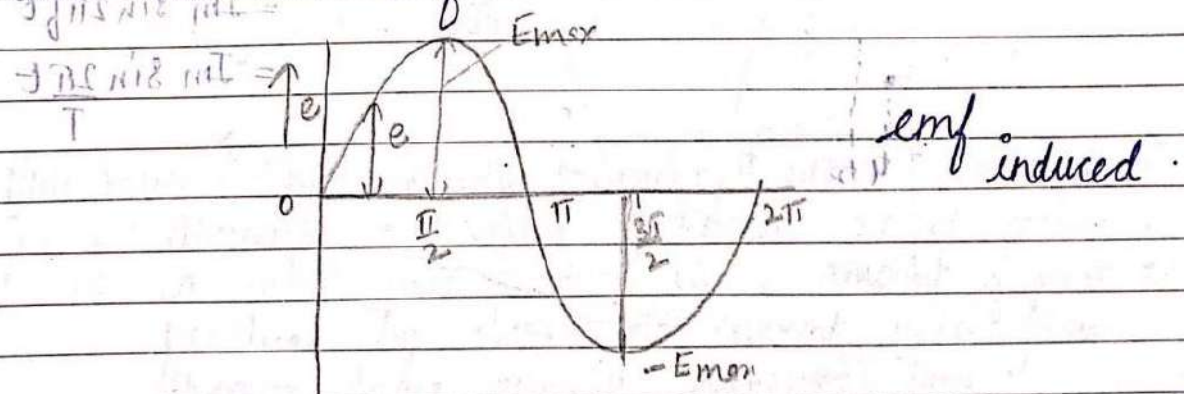
$$\epsilon = - \frac{d}{dt} (N \phi_{max} \cos \omega t)$$

$$= N \phi_{max} \frac{d}{dt} (-\cos \omega t) = N \phi_{max} \sin \omega t \times \omega$$

$$= \omega N \phi_{max} \sin \omega t$$

when $\omega t = 0$ $\sin \omega t = 0$ $\epsilon = 0$
 $\omega t = \pi$ $\sin \pi = 0$ $\epsilon = 0$
 $\omega t = \frac{\pi}{2}$ $\sin \frac{\pi}{2} = 1$ $\epsilon = \epsilon_{max} = \omega N \phi_{max}$
 $\omega t = \frac{3\pi}{2}$ $\sin \frac{3\pi}{2} = -1$ $\epsilon = -\epsilon_{max}$

So ϵ induced varies as the sine function of the time angle ωt . Such ϵ is called the sinusoidal ϵ .



alternating Voltage

$$e = E_{max} \sin \omega t \quad e = E_{max} \sin 2\pi f t \quad e = E_m \sin \frac{2\pi t}{T}$$

alternating Current

$$i = I_{max} \sin \omega t \quad i = I_{max} \sin 2\pi f t \quad i = I_m \sin \frac{2\pi t}{T}$$

7 Peak value, Average value and R.M.S value =

Peak value The max. value attained by alternating quantity during one cycle.

(max value ; crest value) or amplitude =

Average value The arithmetic average of all the instantaneous values considered of an alternating quantity over one cycle.

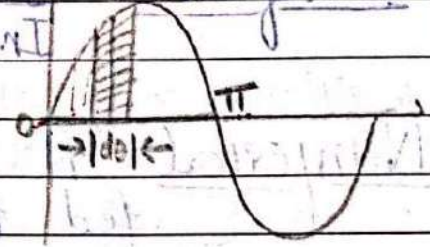
$$I_{av} = \frac{I_1 + I_2 + \dots + I_n}{n} = \frac{\text{Area of alternation}}{\text{base}}$$

$$I_{av} = \frac{1}{\pi} \int_0^{\pi} I_m \sin \theta d\theta$$

$$= \frac{1}{\pi} \left[\int_0^{\pi} I_m \sin \theta d\theta \right]$$

$$= \frac{I_m}{\pi} (-\cos \theta)_0^{\pi} = \frac{I_m}{\pi} (-(-1) - (-1))$$

$$= \frac{I_m}{\pi} (2) = \frac{2I_m}{\pi}$$



Rms Value : That steady current which when flows through a resistor of known resistance for a given time produces same amount of heat as produced by alternating current when flows through same resistor for same time.

$$I_{rms} = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2}{n}}$$

$$I_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} i^2 d\theta}$$

$$I_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \theta d\theta}$$

$$i^2 = (I_m \sin \theta)^2$$

$$I_{rms}^2 \text{ or } (I_{eff})^2 R t = \frac{R t}{n} (i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2)$$

$$= \left[\frac{I_m^2}{\pi} \int_0^\pi \frac{1 - \cos 2\theta}{2} d\theta \right]$$

$$= \frac{I_m^2}{2\pi} \left[(\theta)_0^\pi - \left(\frac{\sin 2\theta}{2} \right)_0^\pi \right]$$

$$= \frac{I_m^2}{2\pi} \left[(\pi - 0) - \left(\frac{\sin 2\pi}{2} - \frac{\sin 2 \cdot 0}{2} \right) \right]$$

$$= \frac{I_m^2}{2\pi} [\pi] = \frac{I_m^2}{2}$$

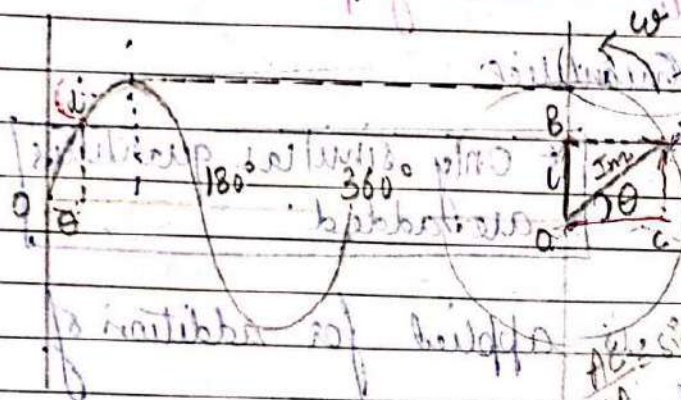
Form factor: $\frac{I_{r.m.s}}{I_{av}} = \frac{I_m/\sqrt{2}}{2I_m/\pi} = \frac{\pi I_m}{2\sqrt{2} I_m} = 1.11$

Peak factor: $\frac{I_m}{I_{r.m.s}} = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2} = 1.4142$

3

Phasor representation of alternating quantities

The alternating quantity can be represented by a line of definite length rotating in counter clockwise direction at const. velocity (ω rad/sec) such rotating line is called phasor.



$$u = I_m \sin \theta$$

OA represent I_m and it is rotating in CCW direction with ω

$$OB = OA \sin \theta = I_m \sin \theta$$

$$OB = i = I_m \sin \theta$$

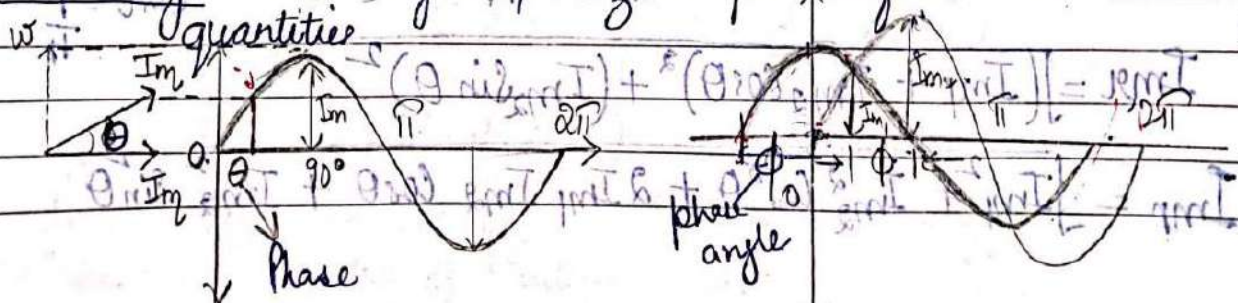
The phasor representation of alternating quantity represent magnitude and position on axis.

Phase and phase difference

Phase:- defined as fractional part of a cycle through which the quantity has advanced from a selected origin. (θ)

Phase difference:- Two alternating quantities having same frequency attain their zero value at different instant.

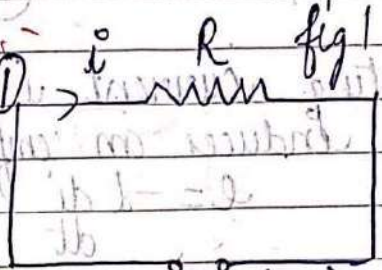
Phase angle The angle b/w zero point of 2 alternating quantities



AC Circuit (with single basic network element)

1. Containing Resistance only

Current
Let voltage applied $V = V_m \sin \omega t$ - (1)



So Current $i = \frac{V}{R}$

$i = \frac{V_m \sin \omega t}{R}$ - (2)

for max current $\omega t = 90^\circ$
 $I_m = \frac{V_m \sin 90^\circ}{R}$ $I_m = \frac{V_m}{R}$

So equation 2 can be written as
 $i = I_m \sin \omega t$ - (3)

Phase angle

from equation 1 and 3 it is clear that there is no phase difference b/w app voltage & current flowing through R.
 phase angle = 0

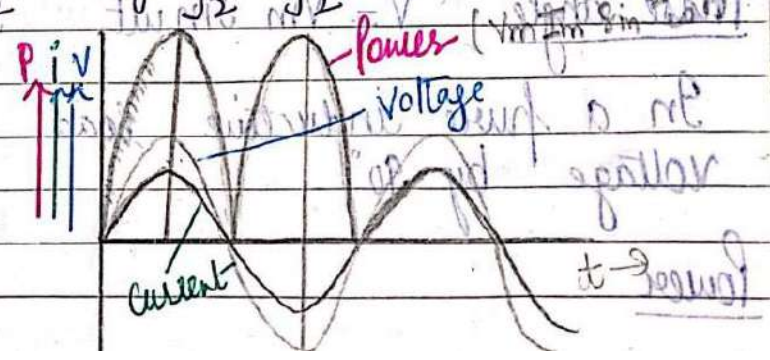
Power

$p = vi = V_m \sin \omega t \times I_m \sin \omega t$
 Instantaneous power
 $= V_m I_m \sin^2 \omega t$
 $= V_m I_m (1 - \cos 2\omega t)$
 $= \frac{V_m \times I_m}{\sqrt{2} \sqrt{2}} (1 - \cos 2\omega t) = \frac{V_m I_m}{2} - \frac{V_m I_m \cos 2\omega t}{2}$

over complete cycle it is zero
 $P = \text{avg. } \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} - \left(\text{avg } \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \omega t \right) = V_{rms} I_{rms} - 0$

$P = VI$

Power curve fig 2.



2. Pure Inductance only.

Let $V = V_m \sin \omega t$ — (1)

Alternating current i flows through the inductance, which induces an emf in it.

$$e = -L \frac{di}{dt}$$

emf is equal & opp. to applied voltage $V = e = -(-L \frac{di}{dt}) = L \frac{di}{dt}$

$$V_m \sin \omega t = L \frac{di}{dt}$$

$$di = \frac{V_m}{L} \sin \omega t dt \quad \int \sin(90 - \omega t) = \cos \theta$$

Integrating both sides —

$$\int di = \int \frac{V_m}{L} \sin \omega t dt$$

$$i = \frac{V_m}{\omega L} (-\cos \omega t)$$

$$i = \frac{V_m}{\omega L} (\cos \omega t)$$

$$i = \frac{V_m}{\omega L} \left(\sin \omega t - \frac{\pi}{2} \right) = \frac{V_m}{\omega L} \left(\sin \omega t - \frac{\pi}{2} \right)$$

$$X_L = \omega L \quad \therefore \omega = 2\pi f \quad \therefore X_L = 2\pi f L \text{ (opposition offered by inductor)}$$

Inductive reactance

Max. current

$$I_m = \frac{V_m}{X_L}$$

$$\left(\sin \omega t - \frac{\pi}{2} \right) = 1$$

$$\text{So } i = I_m \left(\sin \omega t - \frac{\pi}{2} \right) \quad \text{--- (2)}$$

Phase angle

$$V = V_m \sin \omega t$$

$$i = I_m \sin \left(\omega t - \frac{\pi}{2} \right)$$

In a pure inductive load current i lags the voltage by 90°

Power

instantaneous power

$$p = Vi$$

$$= V_m \sin \omega t \times I_m \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$= -V_m I_m \sin \omega t \cos \omega t \times 2$$

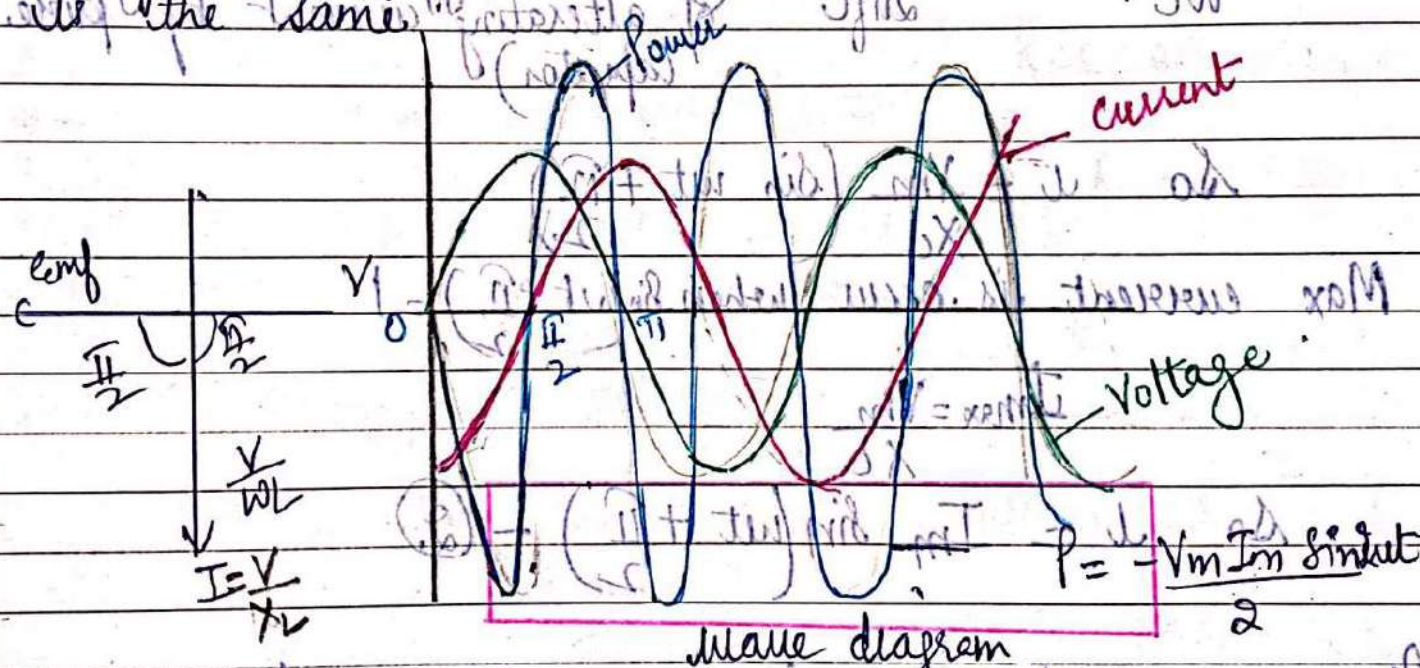
$$= -V_m I_m \sin 2\omega t = -V_m I_m \sin 2\omega t$$

Average power

$$P = \text{avg.} \left(\frac{V_m}{\sqrt{2}} \right) \times \frac{I_m}{\sqrt{2}} \sin 2\omega t = 0$$

Hence no power and energy is consumed in circuit

Power curve. Avg power in half cycle is zero as the negative & positive loop area under power curve is the same.



3. Pure Capacitor circuit

Let $V = V_m \sin \omega t$

Charge stored on capacitor $q = CV$

$$\frac{dq}{dt} = d \cdot i$$

$$q = CV$$

$$\text{So } i = \frac{d(CV)}{dt}$$

$$i = \frac{d(CV_m \sin \omega t)}{dt} = V_m C \frac{d \sin \omega t}{dt}$$

$$i = V_m C \cos \omega t \times \omega$$

$$i = \omega C \times V_m \cos \omega t$$

$$X_C = \frac{1}{\omega C}$$

$$\cos(\omega t) = \sin(\omega t + \frac{\pi}{2})$$

$$i = \frac{V_m}{X_C} \cos \omega t \text{ or } \frac{V_m}{X_C} \sin(\omega t + \frac{\pi}{2})$$

$X_C = \frac{1}{\omega C}$ (Capacitive reactance) is the opposition offered to the flow of alternating current by pure capacitor.

$$\text{So } i = \frac{V_m}{X_C} \sin(\omega t + \frac{\pi}{2})$$

Max current is occur when $\sin(\omega t + \frac{\pi}{2}) = 1$

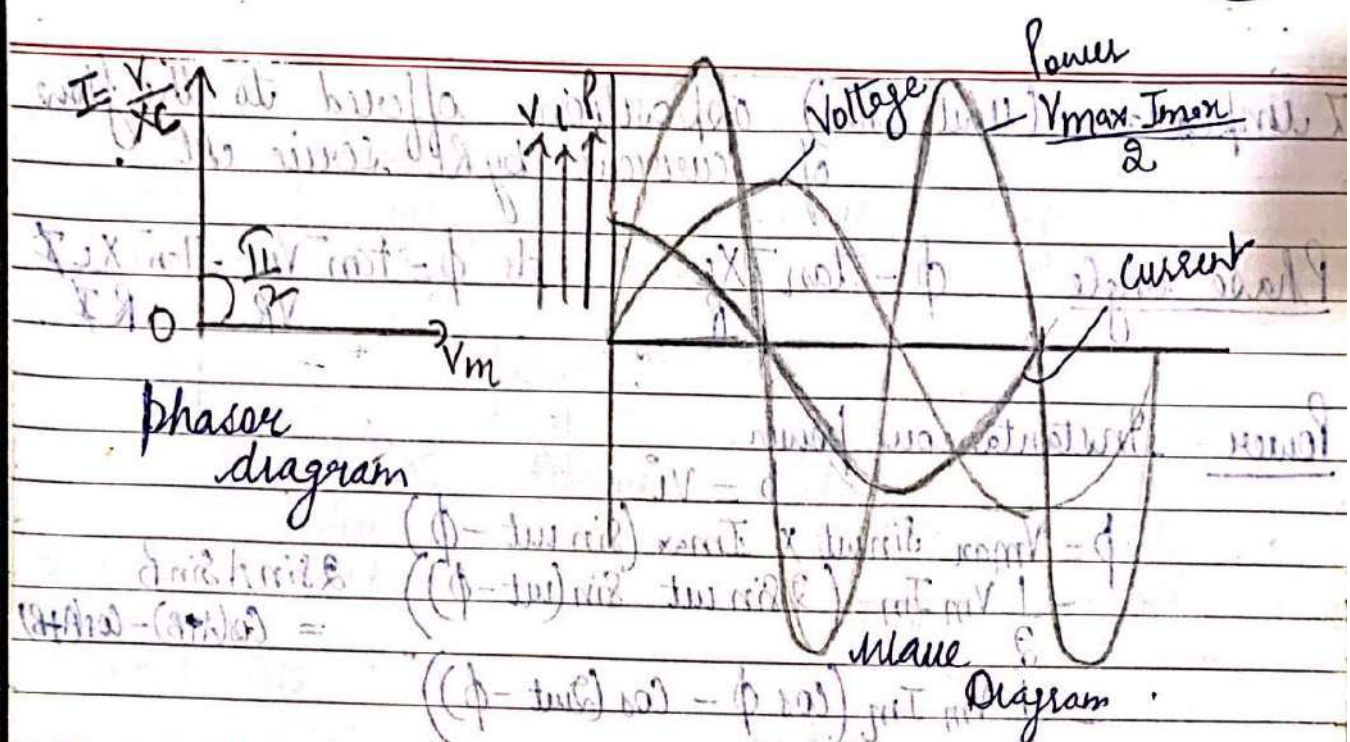
$$i_{\text{max}} = \frac{V_m}{X_C}$$

$$\text{So } i = I_m \sin(\omega t + \frac{\pi}{2}) \quad (2)$$

Phase angle In AC circuit containing pure capacitance; current lead voltage by an angle 90° .

Power $p = VI = V_m \sin \omega t \times I_m \sin(\omega t + \frac{\pi}{2})$
 $= V_m I_m \sin \omega t \cos \omega t$
 $= V_m I_m \sin 2\omega t$

$P_{\text{average}} = 0$ power consumed by pure C is zero



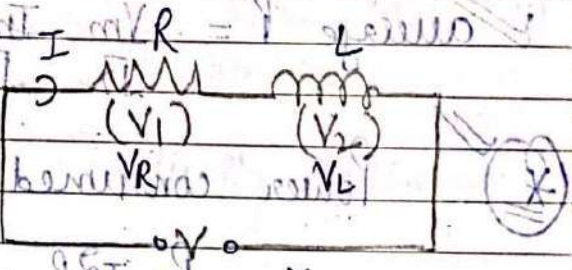
Single phase series circuit

RL Series circuit

RC Series circuit

RLC Series circuits

1. RL circuit AC circuit in which R and L are connected in series.
frequency = f Current = I



Voltage drop across $R = IR$
(in phase with current)

Voltage drop across $L = IX_L = I(\omega L)$
(Lags behind V by 90°)

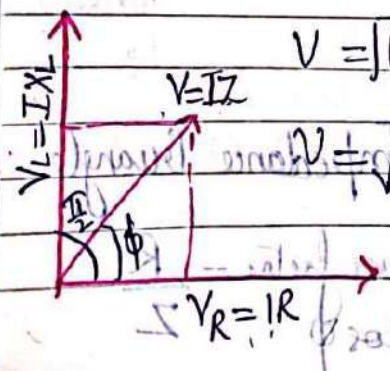
Applied $V = \sqrt{V_R^2 + V_L^2}$ (Ilgen method)

$$V = \sqrt{(IR)^2 + (IX_L)^2}$$

$$V = I \sqrt{R^2 + X_L^2}$$

$$V = I \sqrt{R^2 + X_L^2} \quad V = IZ \quad \text{(ohm)}$$

Impedance



waveform from book.

Impedance (Unit Ohm) opposition offered to the flow of current by RL series ckt.

Phase angle $\phi = \tan^{-1} \frac{X_L}{R}$ As $\phi = \tan^{-1} \frac{V_L}{V_R} = \tan^{-1} \frac{X_L Z}{R Z}$

Power Instantaneous power

$$p = v i$$

$$p = V_{\max} \sin \omega t \times I_{\max} (\sin \omega t - \phi)$$

$$= \frac{1}{2} V_m I_m (2 \sin \omega t \sin(\omega t - \phi))$$

$$= \frac{1}{2} V_m I_m (\cos \phi - \cos(2\omega t - \phi))$$

$$= \frac{V_m I_m}{2} \cos \phi - \frac{V_m I_m}{2} \cos(2\omega t - \phi)$$

for pulsating component

average power = 0

average $P = \frac{1}{\sqrt{2}} V_m \times \frac{1}{\sqrt{2}} I_m \cos \phi = V I \cos \phi$

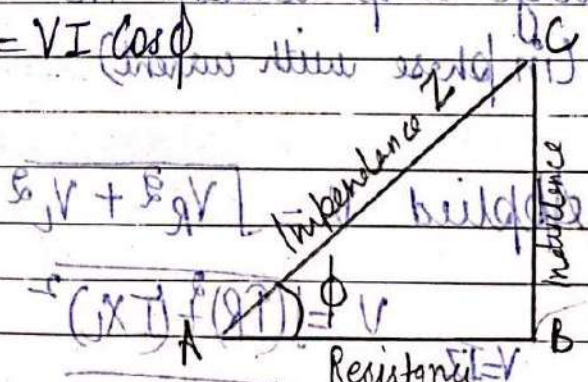
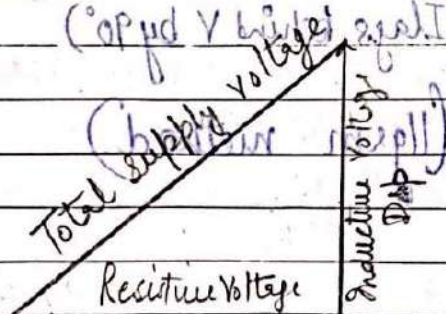
Power consumed by ckt = $I^2 R$ (only due to resistance)

$$P = I^2 R = I (I R) = V (I R)$$

$(I R) = I \cos \phi$

$(I R) = I \cos \phi$

$$P = V I \cos \phi$$



Voltage Triangle

Impedance Triangle

Power factor = $\cos \phi = \frac{R}{Z}$

Power factor

$$\cos \phi = \frac{P}{S}$$

$\cos \phi$ = cosine angle b/w V and I
or $I \cos \phi$

$\cos \phi$ = Ratio of $\frac{R}{Z}$ (Impedance triangle)

$\cos \phi$ = Ratio of True Power / Apparent Power = $\frac{VI \cos \phi}{VI}$ (from Power Triangle).

$\cos \phi < 1$ (it may be leading or lagging)

Active component of current The current component which is in phase with V and contribute to real power is called active component of current or (wattful or inphase) ($I \cos \phi$)

Reactive component of current The current component which is 90° out of phase to circuit V and contribute to reactive power. (wattless) ($I \sin \phi$)

Q factor Reciprocal of power factor is known as Q factor of the coil.

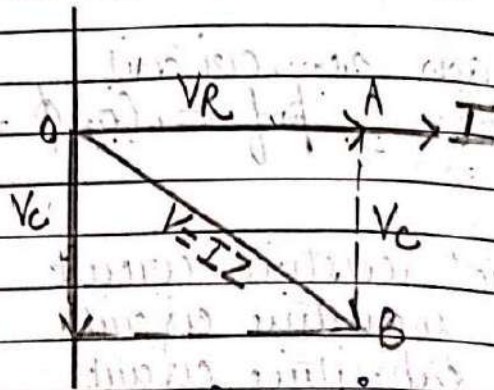
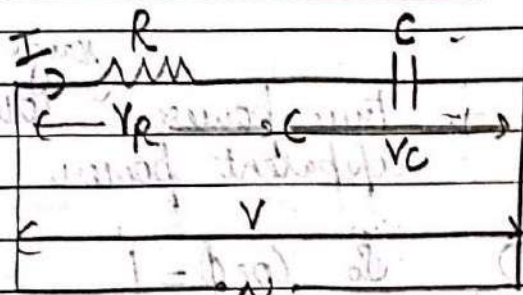
$$Q = \frac{1}{\cos \phi} = \frac{1}{R/Z} = \frac{Z}{R} \quad Q = \sqrt{R^2 + X_L^2}$$

R is small as compared to X_L

$$Q = \frac{X_L}{R} = \frac{\omega L}{R}$$

$$Q = \frac{V}{I R} = \frac{V}{I} \times \frac{1}{R}$$

R-C series circuit



$$V_R = IR \quad V_C = IX_C$$

Phasor diagram

where $X_C = \frac{1}{2\pi fC}$

$$V = \sqrt{(IR)^2 + (IX_C)^2}$$

$$= I \sqrt{R^2 + X_C^2}$$

$$V = IZ \quad \text{where } Z = \sqrt{R^2 + X_C^2}$$

The total opposition offered to flow ac by R-C series circuit

Phase angle

$$\tan \phi = \frac{V_C}{V_R} = \frac{IX_C}{IR} = \frac{X_C}{R}$$

$$\phi = \tan^{-1} \frac{X_C}{R}$$

Power

$$P = VI \cos \phi \quad (\text{due to only pure Resistance})$$

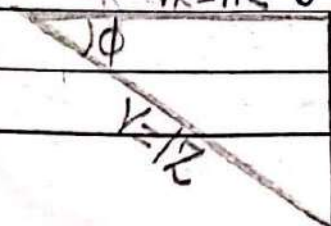
$$= (IZ) I \left(\frac{R}{Z} \right) \quad V = IZ \quad \cos \phi = \frac{R}{Z}$$

$$P = I^2 R$$

power curve

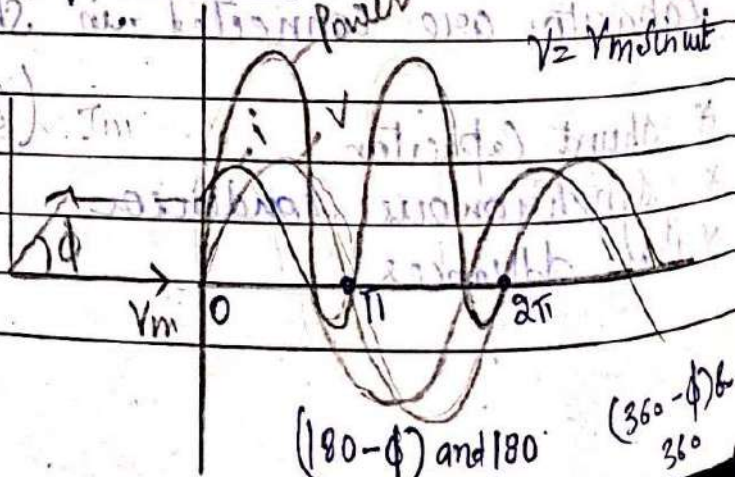
Impedance triangle

$$V_R = IR$$

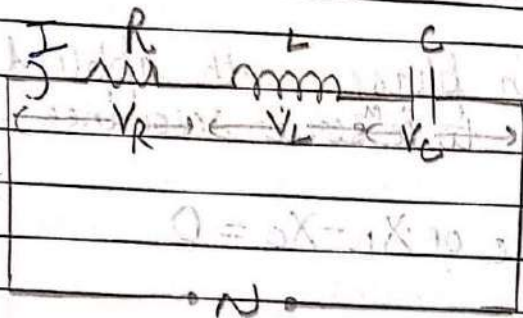


$$V_C = IX_C$$

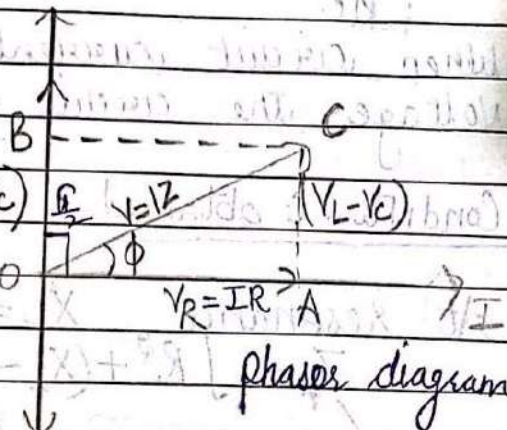
$$Z = IZ$$



RLC series circuit



$$V_L = IX_L$$



When current I flows through a circuit, the voltage across each component will be

$$V_R = IR$$

(in phase)

$$V_L = IX_L$$

V leads I by 90°

$$V_C = IX_C$$

V lags I by 90°

$$\text{So } V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$\text{or } V = I \sqrt{R^2 + (X_L - X_C)^2} \text{ or } V = IZ$$

where $Z = \sqrt{R^2 + (X_L - X_C)^2}$ opposition offered to I by RLC series circuit.

Phase angle $\tan \phi = \frac{V_L - V_C}{V_R} = \frac{IX_L - IX_C}{IR} = \frac{X_L - X_C}{R}$

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

Power: Average Power = $I^2 R$ P.f. $\cos \phi = \frac{R}{Z} = \frac{V_R}{V}$

Voltage $V_m = V_m \sin \omega t$

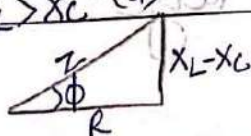
Current $i = I_m \sin(\omega t - \phi)$ $X_L > X_C$

$i = I_m \sin(\omega t + \phi)$ $X_L < X_C$

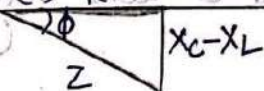
$i = I_m \sin \omega t$ $X_L = X_C$

Impedance Δ

when $X_L > X_C$ (a)



(b) $X_L < X_C$



Series Resonance

in RLC
When circuit current is in phase with applied voltage the circuit is said to be in series resonance.

Condition is obtained $X_L = X_C$ or $X_L - X_C = 0$

At resonance $X_L = X_C$

$$Z_r = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2} = R$$

Resonance impedance $V = I Z_r$ $V = I Z_p$

$$I_r \text{ (resonance current)} = \frac{V}{Z_r} = \frac{V}{R}$$

$$V = I Z$$

* At resonance the opposition to flow of current is only due to R , hence circuit draws max. current.

Resonant frequency $X_L = X_C$

$$2\pi f L = \frac{1}{2\pi f C}$$

$$f_r^2 = \frac{1}{4\pi^2 L C}$$

$$f_r = \frac{1}{2\pi \sqrt{LC}}$$

Effect of series resonance

1. $Z_r = R$ min. impedance.

2. $I_r = \frac{V}{Z_r}$ max. current.

3. $P = I_r^2 R$ max. power

4. $I_r \uparrow$ V.D. across L and C also very large.

* In power system.

At resonance, the excessive voltage build up across the inductive and capacitive component (CB, reactance etc). This may cause damage these components. $\therefore$ series resonance should be avoided in power system.

In electronic devices such as antenna circuit of radio and TV, Tuning circuits etc, the principle of series resonance is used to

As $\uparrow$ the signal voltage & current at desired frequency (f_r)

A series resonant circuit has capability to draw heavy current & power from the mains, so it is also known as acceptor circuit.

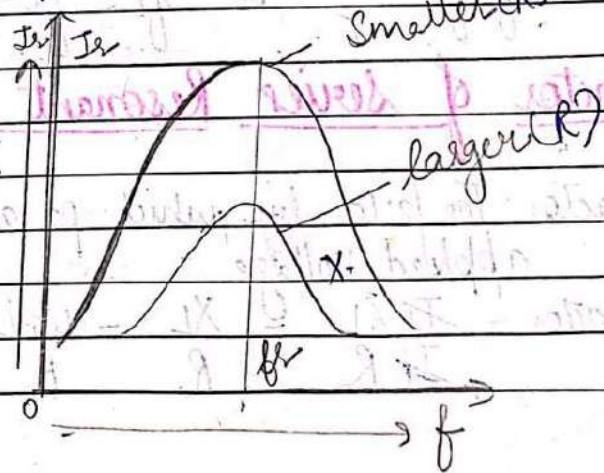
Resonance curve is plotted when the graph b/w current & frequency the curve which obtained is known as resonance curve.

Resonance curve of RLC series circuit

$$f < f_r \quad X_c > X_L$$

$$f > f_r \quad X_L > X_c$$

$$Z > Z_r$$



* I_r is max current at f_r .

* $f < f_r$ $X_c > X_L$ * In both the

$f > f_r$ $X_c < X_L$

Cases $Z > Z_r$

so the value of current decreases

$$V = IZ$$

$$(less) I = \frac{V}{Z}$$

(more)

This is the reason why current falling off rapidly on the both side

* Smaller the R greater the current at resonance

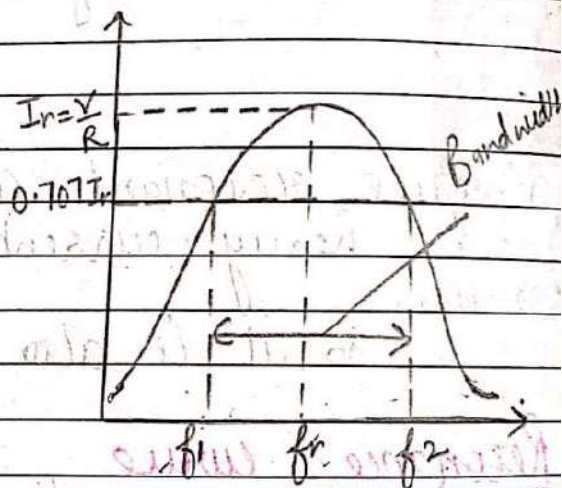
Bandwidth

The range of f over which current is equal to more than 70.7% of I_r is known as bandwidth

$$BW = f_2 - f_1$$

higher or upper cut off f

lower cut off



$$* f_1 = \sqrt{f_1 f_2}$$

$$* f_1 = f_r - \frac{BW}{2}$$

$$* f_2 = f_r + \frac{BW}{2}$$

Q factor of series Resonant Circuit

Q factor: The factor by which pd across L or C rises to that of applied voltage

$$Q \text{ factor} = \frac{I_r X_L}{I_r R} = \frac{X_L}{R} = \frac{\omega_r L}{R} = \frac{2\pi f_r L}{R} = \frac{2\pi X_L}{2\pi \sqrt{L/C} R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

AC Parallel circuit (extra)

The ac circuit in which no. of branches are connected in such a way that V across each branch is same but current flowing through them is different.

Methods for solving parallel ckt.

Phasor or

Vector method

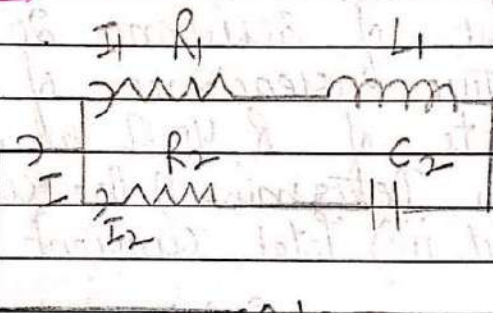
Admittance method

method of phasor algebra or
I method.

1. Phasor or vector method

Step 1st find impedance of each branch

$$Z_1 = \sqrt{R_1^2 + X_L^2} \quad Z_2 = \sqrt{R_2^2 + X_C^2}$$

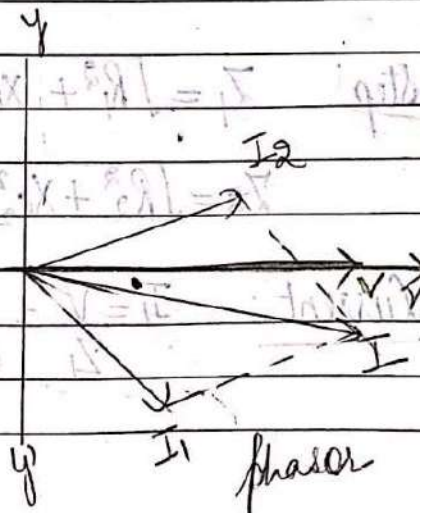


Step 2nd Determine the ^{magnitude of} current & phase angle with voltage in each branch.

$$I_1 = \frac{V}{Z_1} \quad \phi_1 = \tan^{-1} \frac{X_L}{R_1} \text{ (lagging)}$$

$$I_2 = \frac{V}{Z_2} \quad \phi_2 = \tan^{-1} \frac{X_C}{R_2} \text{ (leading)}$$

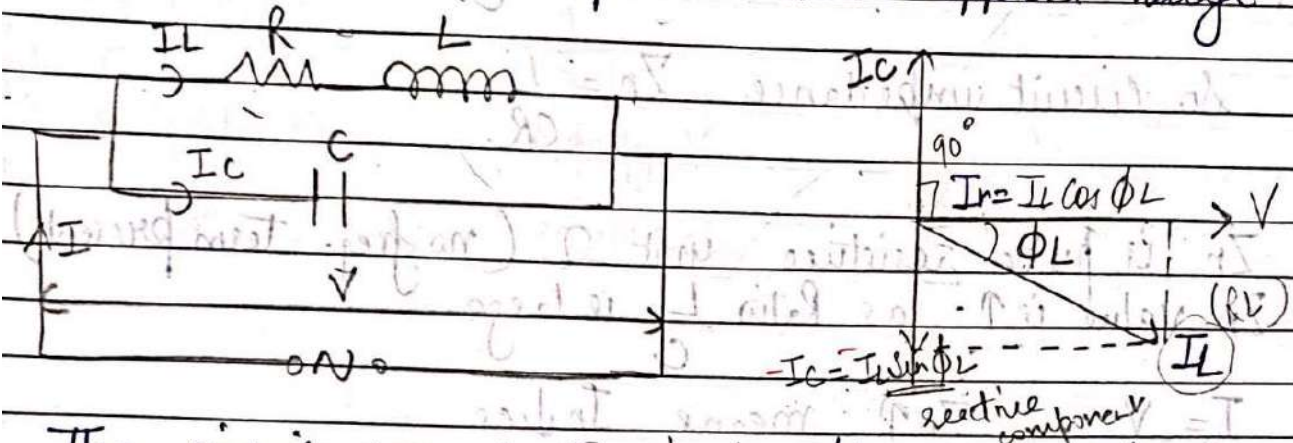
Step 3rd Draw the phasor diagram taking Voltage as reference phasor.



9

Parallel resonance

An AC circuit containing an inductor & C in parallel is said to be in parallel resonance when the current is in phase with applied voltage.

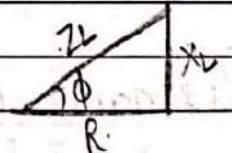


The circuit current I_r is in phase with V
 $I_C = I_L \sin \phi_L$

At resonance reactive component of current is suppressed, the circuit draws min. current under this condition.

Resonant freq.

At parallel resonance
 $I_C = I_L \sin \phi_L$



$$\frac{I_L = \frac{V}{Z_L}}{I_C = \frac{V}{X_C}} \quad \sin \phi_L = \frac{X_L}{Z_L} \quad \text{[Circuit]}$$

$$\frac{V}{X_C} = \frac{V}{Z_L} \times \frac{X_L}{Z_L}$$

$$Z_L^2 = X_L X_C$$

$$f_r = \frac{1}{2\pi \sqrt{LC}}$$

$$Z_L^2 = R^2 + X_L^2 = \frac{WL}{\omega C}$$

$$R^2 + 4\pi^2 f_r^2 L^2 = \frac{L}{C}$$

$$4\pi^2 f_r^2 L^2 = \frac{L}{C} - R^2$$

$$f_r^2 = \frac{1}{4\pi^2 L^2} \left(\frac{L}{C} - R^2 \right) = \frac{1}{4\pi^2 LC} \left(1 - \frac{R^2 C}{L} \right)$$

$$f_r = \frac{1}{2\pi \sqrt{LC}} \sqrt{1 - \frac{R^2 C}{L}}$$

Effect of parallel resonance

line current

$$Z_L^2 = \frac{L}{C} \text{ placed above}$$

$$I_r = I_L \cos \phi$$

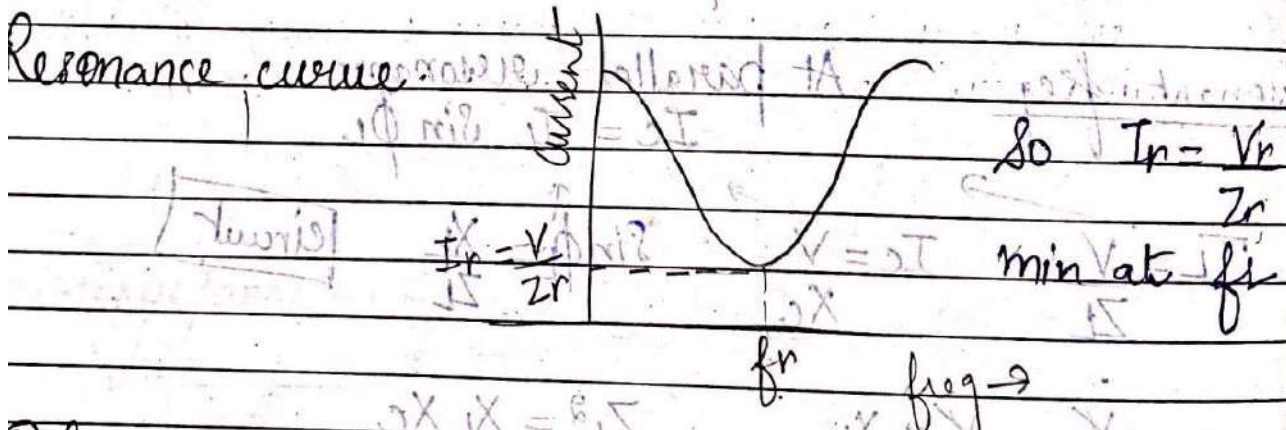
$$V = V \times \frac{R}{Z_L} ; \frac{1}{Z_r} = \frac{R}{Z_L^2} ; \frac{1}{Z_r} = \frac{R}{\frac{L}{C}} = \frac{CR}{L}$$

So circuit impedance $Z_r = \frac{L}{CR}$

1. Z_r is pure resistive unit Ω (no freq. term present)
2. Z_r value is $\uparrow$ as Ratio $\frac{L}{C}$ is large.

$$3. I_r = \frac{V}{Z_r} \quad Z_r \uparrow \text{ means } I_r \text{ decreases}$$

Since in parallel resonant ckt it draws a very small current and power so it is often regarded as rejector ckt.



Q factor

$$Q = \frac{I_{C_{max}}}{I_r} = \frac{V/X_C}{V/Z_r} = \frac{Z_r}{X_C} = \frac{L}{CR} \times 2\pi f_r L = \frac{2\pi f_r L}{R}$$

$$Q = \frac{2\pi L}{R} \cdot \frac{1}{2\pi LC} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

* Comparison Series & Parallel resonance.

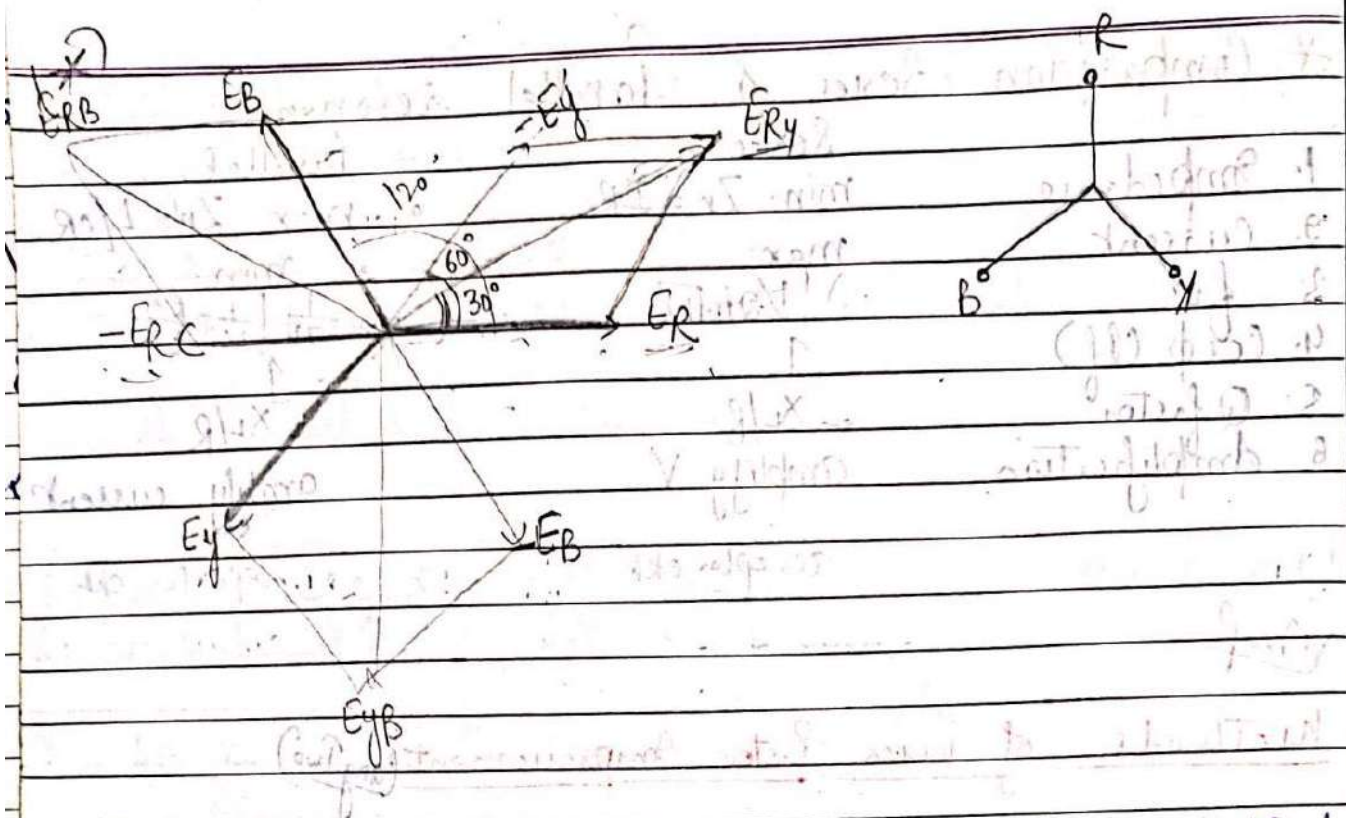
	Series	Parallel
1. Impedance	min. $Z_r = R$	max $Z_p = L/CR$
2. Current	max	min
3. f_r	$V/2\pi LC$	$\frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$
4. $\cos \phi$ (p.f.)	1	1
5. Q factor	X_L/R	X_L/R
6. Amplification	amplify V	amplify current
	acceptor ckt.	rejector ckt.

V.M.F

Methods of Power Factor Improvement (any two)

1. Capacitor: Improving power factor means reducing the phase difference between Voltage & Current. Since the majority of loads are of Inductive nature, they require some amount of reactive power for them to function. The Capacitors are installed in parallel to the load provided. This reactive power it They act as source of local reactive power, and thus less reactive power flows through the line. This reduce the phase difference b/w the V & I.

2. Synchronous Condenser 3 ϕ Synchronous motor with no load attached to its shaft. The syno. motor has characteristic of operating under any p.f. leading, lagging or unity depending upon the excitation. For inductive loads, a synchronous condenser is connected towards load side and is overexcited. Synchronous condenser makes it behave like a capacitor. It draws the lagging current from the supply or supplies the reactive power.



3. Phase advancer This is an ac exciter mainly used to improve pf of Induction motor. They are mounted on the shaft of motor and connected to the rotor circuit of motor. It improves the pf by providing the exciting ampere turns to produce required flux at slip frequency. Further, if ampere turn per ϕ it can be made to operate at leading power factor.

Polyphase s/s. (poly = many)

AC system having a group of 2 or more, equal voltages of same freq. arranged to have equal phase difference b/w them.

$$\text{phase difference} = \frac{360^\circ}{\text{No. of phases.}}$$

Unbalance source

3 ϕ supply will be unbalance when either of 3 ϕ voltages are unequal in magnitude as well as in phase.

Unbalance load.

Any 3 ϕ load in which the impedance in one or more phases differ from the impedance of other phases is called an unbalance 3 ϕ load.

Advantages of 3 ϕ system over 1 ϕ system

1. Constant power (ie the total power delivered is constant if load are balanced.)

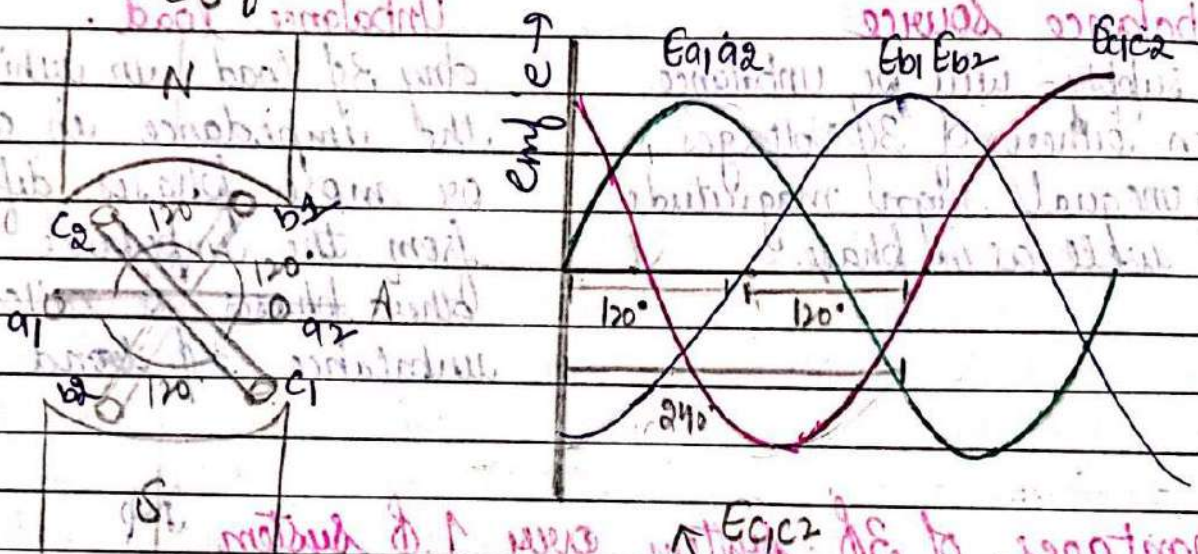
(As in 1 ϕ I and V are in same phase, so power is zero twice in each cycle. When I lead or lags behind V power is -ive twice and zero four time which cause vibration in large motor.)

2. Rating of machine $\uparrow$ w/ increase in no. of phase
So higher o/p in 3 ϕ it is 1.5 times the o/p of 1 ϕ

3. 3 ϕ motor are self starting while 1 ϕ induction motor have no starting torque so need of providing an auxiliary mean for starting.

4. 3 ϕ motor have high pf & η
5. less cost : (3 ϕ s/s require $\frac{3}{4}$ wt of Copper of that required by 1 ϕ to transmit same power at given V & over a give distance)

Generation of EMFs.



$$e_{a1a2} = E_{\max} \sin \omega t$$

$$e_{b1b2} = E_{\max} \sin (\omega t - 120^\circ)$$

$$e_{c1c2} = E_{\max} \sin (\omega t - 240^\circ)$$

Or

$$= E_{\max} \sin (\omega t + 120^\circ)$$

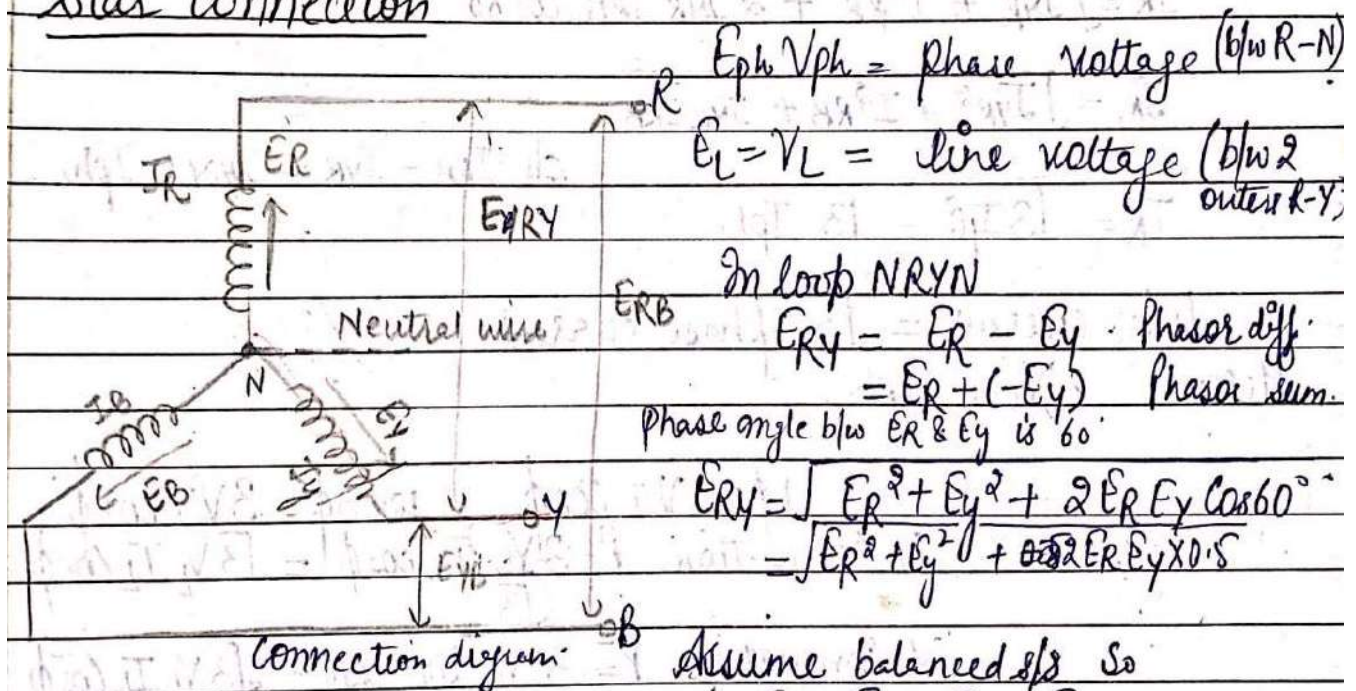
Phase Sequence

The order in which the the current & voltage in 3ϕ attain their maximum value one after other. (abc)

Importance 1. The direction of rotation of 3ϕ IM depend upon the phase sequence of 3ϕ supply. To reverse the direction of rotation, the phase sequence of supply given to the motor has to be changed.

2. The parallel operation of 3ϕ alternator & T/F is only possible if phase sequence is known.

Star Connection



Assume balanced s/s So

$$E_R = E_Y = E_B = E_{ph}$$

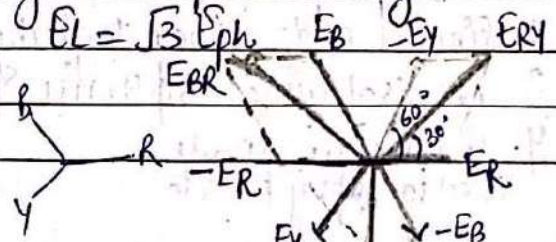
$$E_{RY} = \sqrt{3} E_{ph} = \sqrt{3} E_{ph}$$

$$\text{line voltage} = \sqrt{3} \text{ phase voltage}$$

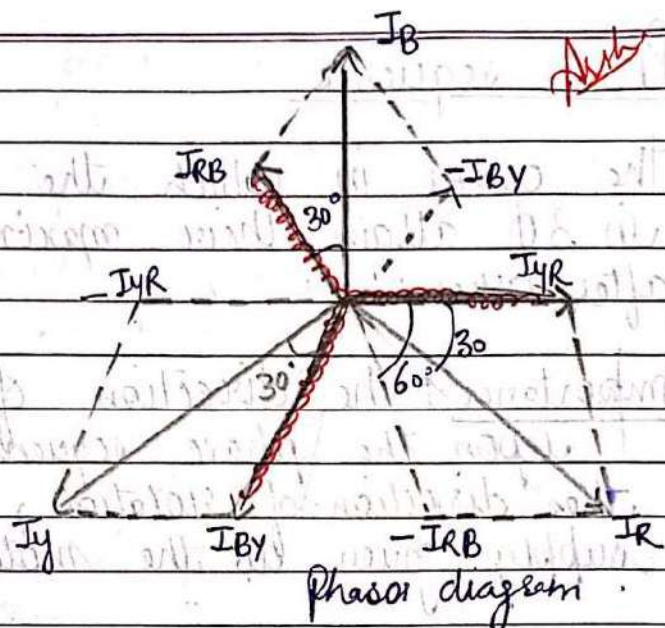
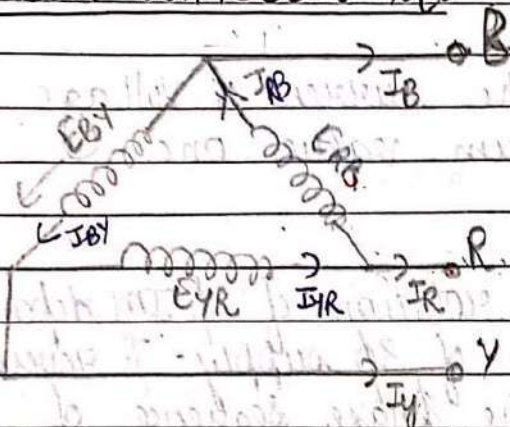
$$\therefore E_L = \sqrt{3} E_{ph}$$

* Same amount of current flow through the phase windg as well as line conductor

$$\text{So } I_L = I_{ph}$$



Delta connected s/s.



line current

$$I_R = I_{YR} - I_{RB}$$

$$= I_{YR} + (-I_{RB})$$

phase angle b/w these current is 60°

$$I_R = \sqrt{I_{YR}^2 + I_{RB}^2 + 2 I_{YR} I_{RB} \cos 60^\circ}$$

$$I_R = \sqrt{I_{YR}^2 + I_{RB}^2 + I_{YR} I_{RB}}$$

$$\text{As } I_{RB} = I_{YR} = I_{BY} = I_{ph}$$

$$I_R = \sqrt{3 I_{ph}^2} = \sqrt{3} I_{ph}$$

line current = $\sqrt{3}$ phase current

Here $V_L = V_{ph}$

Power in 3 ϕ ckt

$$1 \phi \text{ is } V I \cos \phi \quad \text{For } 3 \phi = 3 V I \cos \phi$$

$$\Delta \text{ Connection } P = 3 \frac{V_L I_L}{\sqrt{3}} \cos \phi = \sqrt{3} V_L I_L \cos \phi$$

$$Y \text{ Connection } P = 3 \frac{V_L I_L}{\sqrt{3}} \cos \phi = \sqrt{3} V_L I_L \cos \phi$$

Comparison Star s/s

1. $V_L = \sqrt{3} V_{ph}$ $I_L = I_{ph}$
2. Similar ends joined together
3. Neutral wire 3 ϕ 4 wire s/s
4. Domestic loads used to supply power to

Delta s/s

1. $V_L = V_{ph}$ $I_L = \sqrt{3} I_{ph}$
2. dissimilar ends joint together
3. No neutral wire
4. used in transformer for power transmission